

4. class, Electrodynamics and Optics

Page 688, Task 40.

(a) Two spheres have radii a and b , and their centers are a distance d apart. Show that the capacitance of this system is

$$C = \frac{4\pi\epsilon_0}{\frac{1}{a} + \frac{1}{b} - \frac{2}{d}}$$

provided d is large compared with a and b . Suggestion: Because the spheres are far apart, assume the potential of each equals the sum of the potentials due to each sphere.

(b) Show that as d approaches infinity, the above result reduces to that of two spherical capacitors in series.

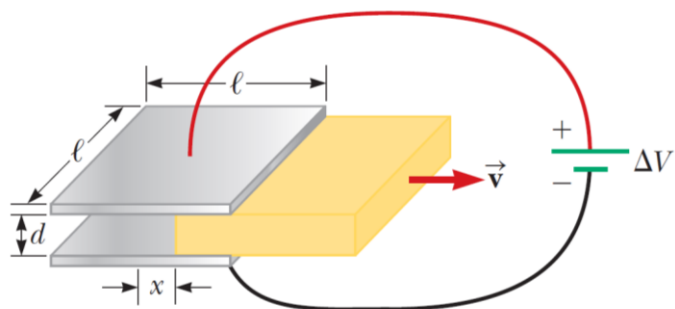
Page 689, Task 46.

Consider two long, parallel, and oppositely charged wires of radius r with their centers separated by a distance D that is much larger than r . Assuming the charge is distributed uniformly on the surface of each wire, show that the capacitance per unit length of this pair of wires is

$$\frac{C}{l} = \frac{\pi\epsilon_0}{\ln(D/r)}$$

Page 712, Task 45.

A parallel-plate capacitor consists of square plates of edge length l , that are separated by a distance d , where $d \ll l$. A potential difference ΔV is maintained between the plates. A material of dielectric constant κ fills half the space between the plates. The dielectric slab is withdrawn from the capacitor as shown in the figure.

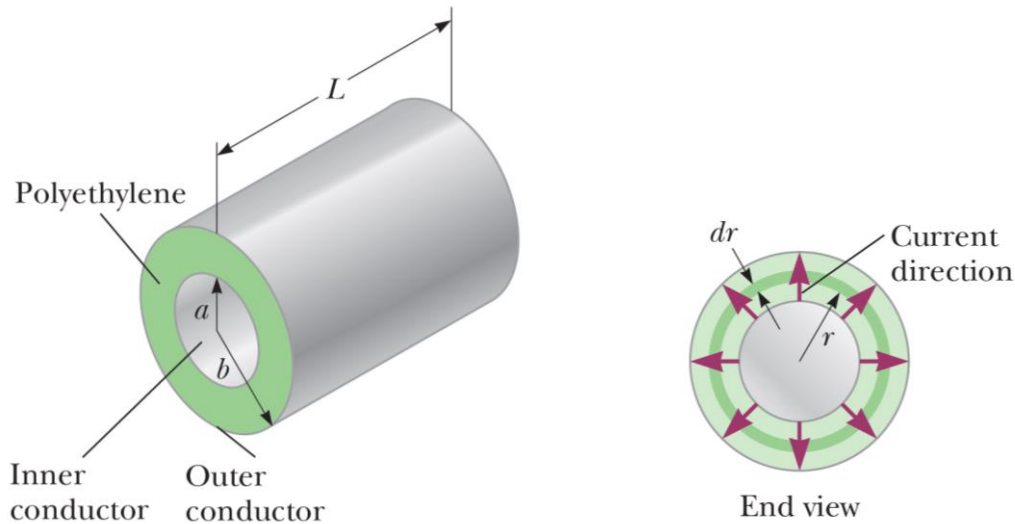


(a) Find the capacitance when the left edge of the dielectric is at a distance x from the center of the capacitor.

(b) If the dielectric is removed at a constant speed v , what is the current in the circuit as the dielectric is being withdrawn?

Page 698, Task 26.3.

Coaxial cables are used extensively for cable television and other electronic applications. A coaxial cable consists of two concentric cylindrical conductors. The region between the conductors is completely filled with polyethylene plastic as shown in the figure. Current leakage through the plastic, in the radial direction, is unwanted. (The cable is designed to conduct current along its length, but that is not the current being considered here.) The radius of the inner conductor is $a = 0.50$ cm, the radius of the outer conductor is $b = 1.75$ cm, and the length is $L = 15.0$ cm. The resistivity of the plastic is $1.0 \cdot 10^{13} \Omega m$. Calculate the radial resistance of the plastic between the two conductors.



Page 709, Task 6.

The figure represents a section of a conductor of nonuniform diameter carrying a current of $I = 5.00$ A. The radius of cross-section A_1 is $r_1 = 0.400$ cm.

(a) What is the magnitude of the current density across A_1 ?

The radius r_2 at A_2 is larger than the radius r_1 at A_1 . (b) Is the current at A_2 larger, smaller, or the same?

(c) Is the current density at A_2 larger, smaller, or the same?

Assume $A_2 = 4 \cdot A_1$. Specify the (d) radius, (e) current, and (f) current density at A_2 .

